

Chapter 14: Light and the Formation of Shadows

Textbook Exercise Solutions & Answers



1. Choose an appropriate word and fill in the blanks.

(a) A is a natural source of light.

star

(b) A is an artificial source of light.

candle

(c) When light passes through a prism, it gets separated into colours.

seven

(d) The image obtained in the pinhole camera is

inverted

(e) A shadow is formed when an object comes in the way of light.

opaque

(f) When a object comes in the way of light, light passes it. options : seven, star, through, transparent, opaque, colors, shape, erect, inverted, luminous, candle.

transparent, through

2. Write whether the following objects are luminous or non-luminous.

Object	Luminous /Non-luminous
A book	Non-luminous
A burning candle	Luminous
A wax cloth	Non-luminous
A pencil	Non-luminous
A pen	Non-luminous
A light bulb	Luminous
A tyre	Non-luminous
A torch	Luminous
Stars	Luminous
The planets	Non-luminous
A satellite	Non-luminous
The Moon	Non-luminous

3. Match the following.

Natural sources of light Man-made sources of light The Sun Tubelight Stars in the night sky Light bulb Fireflies Torch Anglerfish Burning candle Honey mushroom Oil lamps

Transparent	Opaque	Translucent
Piece of glass Water	White plastic Tea kettle Note book Cloth Wooden cupboard	Tinted glass Oil paper Sheet of notebook Wax paper

4. Write the answers to the following.

(a) What things are necessary for the formation of a shadow?

Things necessary for the formation of a shadow are:

1. A source of light
2. An object
3. A surface or screen on which the shadow is formed

(b) When can an object be seen?

We can see an object when reflected rays reach our eyes.

(c) What is a shadow?

1. If an opaque object comes in the way of a light source, light does not pass through it.
2. As a result the light does not reach a wall or any other surface on the other side of the object.
3. That part remains dark. This dark part is called the shadow of the object.